Group Members

Runyan Xin, SeongJun Kwon

Tentative Title

Estimating the Causal Effect of Delivery Delay on Customer Ratings in E-commerce

Problem Description

This project aims to estimate the causal effect of delivery delay on customer ratings using a real-world e-commerce dataset containing approximately 120,000 orders and over 40 variables.

Since delivery delays are not randomly assigned, the main challenge lies in addressing confounding bias. To achieve causal identification, we assume unconfoundedness, meaning that after controlling for observed covariates (e.g., delivery distance, shipping cost, and product price), the treatment assignment is independent of potential outcomes. We also assume overlap, ensuring that for each covariate profile, there is a positive probability of both delayed and non-delayed deliveries.

Under these assumptions, our goal is to quantify the causal impact of delivery delay on customer satisfaction, as measured by ratings.

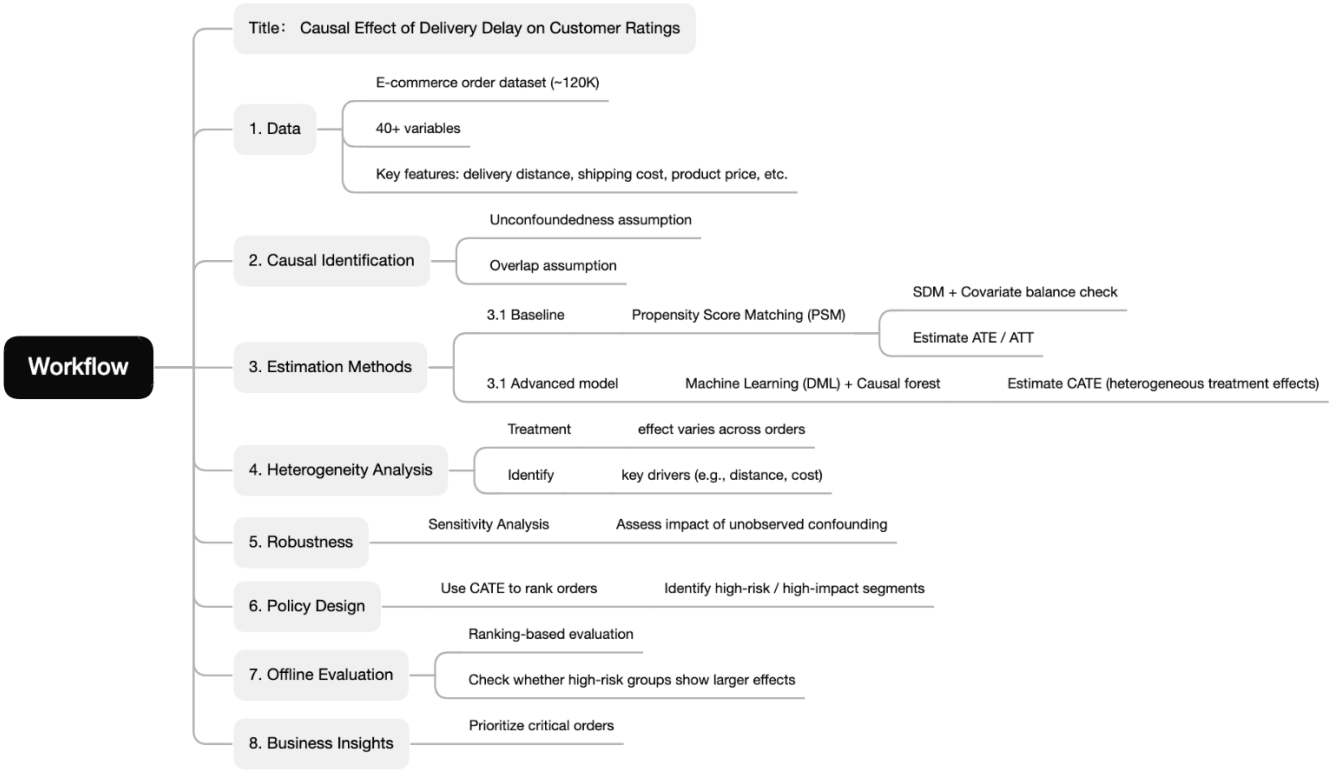
Proposed Plan

We will begin with data preprocessing, including cleaning, feature engineering, and selection of relevant covariates. Potential confounders will be identified based on domain knowledge and exploratory analysis.

As a baseline, we will implement Propensity Score Matching (PSM) to estimate average treatment effects (ATE/ATT) and evaluate covariate balance. We will then apply Double Machine Learning (DML) and Causal Forest methods to estimate heterogeneous treatment effects (CATE), allowing us to understand how the impact of delivery delay varies across different types of orders.

We will also conduct sensitivity analysis to assess the robustness of our findings.

Finally, based on the estimated CATE, we will design a simple targeting strategy that prioritizes orders most sensitive to delivery delays. Since the analysis is based on observational offline data, we will evaluate this strategy using a ranking-based approach to examine whether high-risk groups exhibit larger observed effects.


Workflow
Title： Causal Effect of Delivery Delay on Customer Ratings
1. Data
E-commerce order dataset (~120K)
40+ variables
Key features: delivery distance, shipping cost, product price, etc.
2. Causal Identification
Unconfoundedness assumption
Overlap assumption
3. Estimation Methods
3.1 Baseline
Propensity Score Matching (PSM)
SDM + Covariate balance check
Estimate ATE / ATT
3.1 Advanced model
Machine Learning (DML) + Causal forest
Estimate CATE (heterogeneous treatment effects)
4. Heterogeneity Analysis
Treatment
effect varies across orders
Identify
key drivers (e.g., distance, cost)
5. Robustness
Sensitivity Analysis
Assess impact of unobserved confounding
6. Policy Design
Use CATE to rank orders
Identify high-risk / high-impact segments
7. Offline Evaluation
Ranking-based evaluation
Check whether high-risk groups show larger effects
8. Business Insights
Prioritize critical orders